

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: FPGA Based System Design (01CT0718)	Aim: To design and implement FIR/IIR filters using programmable hardware architectures.	
Experiment No: 13	Date:	Enrolment No: 92301733025

Aim: To design and implement FIR/IIR filters using Verilog HDL on the Xilinx Artix-7 FPGA Board.

Software required:

- Xilinx Vivado Design Suite
- Verilog HDL
- Vivado Simulator

Hardware required:

- Digilent Nexys 4 DDR Development Board
- Xilinx Artix-7 FPGA (XC7A100T-1CSG324C)
- USB programming cable
- Computer/Laptop

Theory:

Digital filters are used to process digital signals by removing unwanted frequency components or modifying the characteristics of the input signal.

The two main types of digital filters are:

1. FIR Filter

FIR (Finite Impulse Response) filters use only present and previous input samples.

The general FIR equation is:

$$y[n] = \sum_{k=0}^{N-1} h[k]x[n-k]$$

For a 4-tap FIR filter:

$$y[n] = h_0x[n] + h_1x[n-1] + h_2x[n-2] + h_3x[n-3]$$

FIR filters do not use feedback and are inherently stable.

2. IIR Filter

IIR (Infinite Impulse Response) filters use both previous input samples and previous output samples.

A general IIR equation is:

$$y[n] = b_0x[n] + b_1x[n-1] - a_1y[n-1]$$

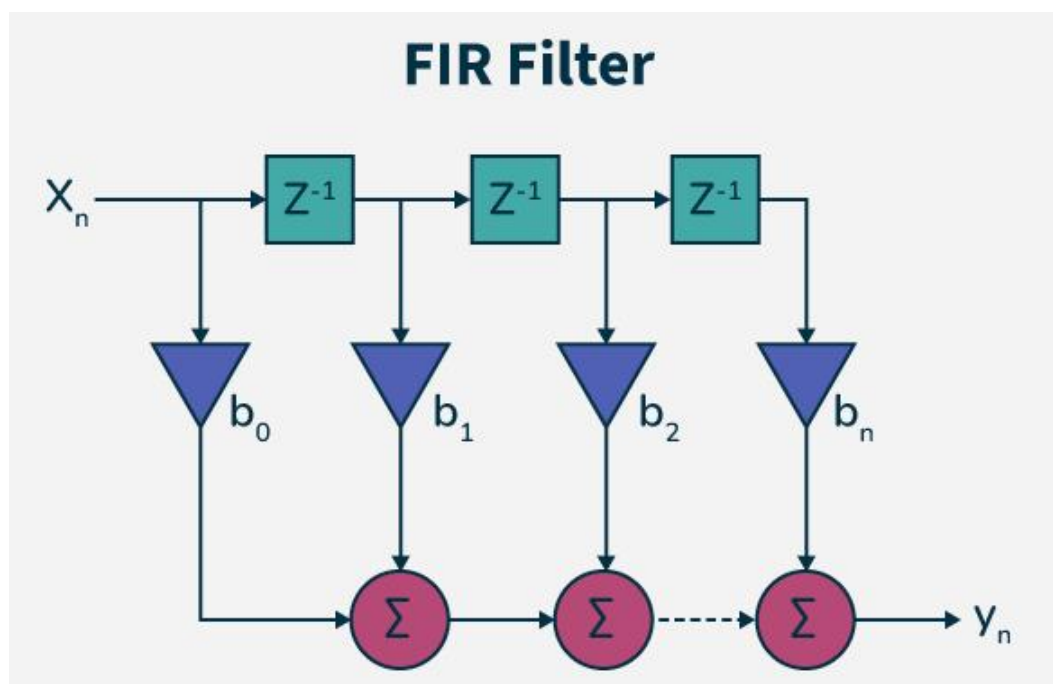
IIR filters use feedback, allowing similar filtering characteristics with fewer coefficients than FIR filters.

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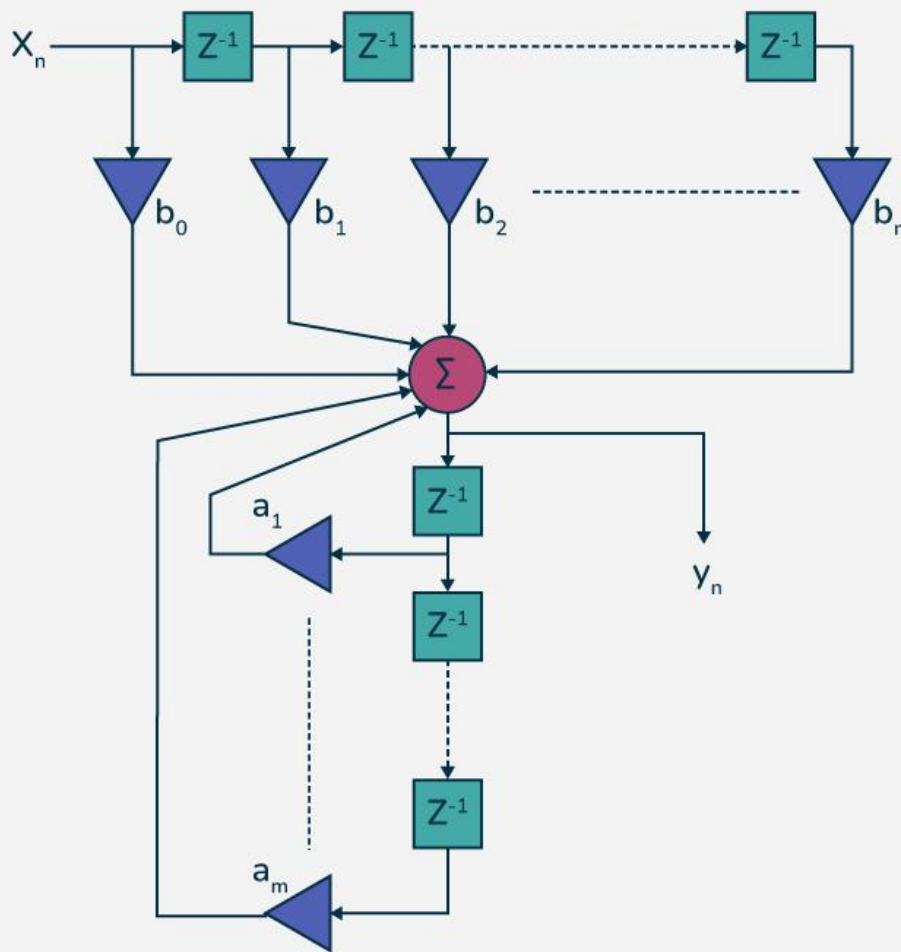
Comparison Between FIR and IIR

Parameter	FIR Filter	IIR Filter
Feedback	No	Yes
Stability	Always stable	Depends on coefficients
Phase Response	Can provide linear phase	Generally non-linear
Hardware	More multipliers for higher order	Fewer resources
Memory	Input samples	Input and output samples

Block diagram:



IIR filter



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Current Input	Previous Input 1	Previous Input 2	Previous Input 3	Output
$x[n]$	$x[n-1]$	$x[n-2]$	$x[n-3]$	$y[n]$
1	0	0	0	1
1	1	0	0	2
1	1	1	0	3
1	1	1	1	4

Boolean Expression:

FIR Filter

$$y[n] = h_0x[n] + h_1x[n-1] + h_2x[n-2] + h_3x[n-3]$$

For the selected coefficients:

$$y[n] = x[n] + x[n-1] + x[n-2] + x[n-3]$$

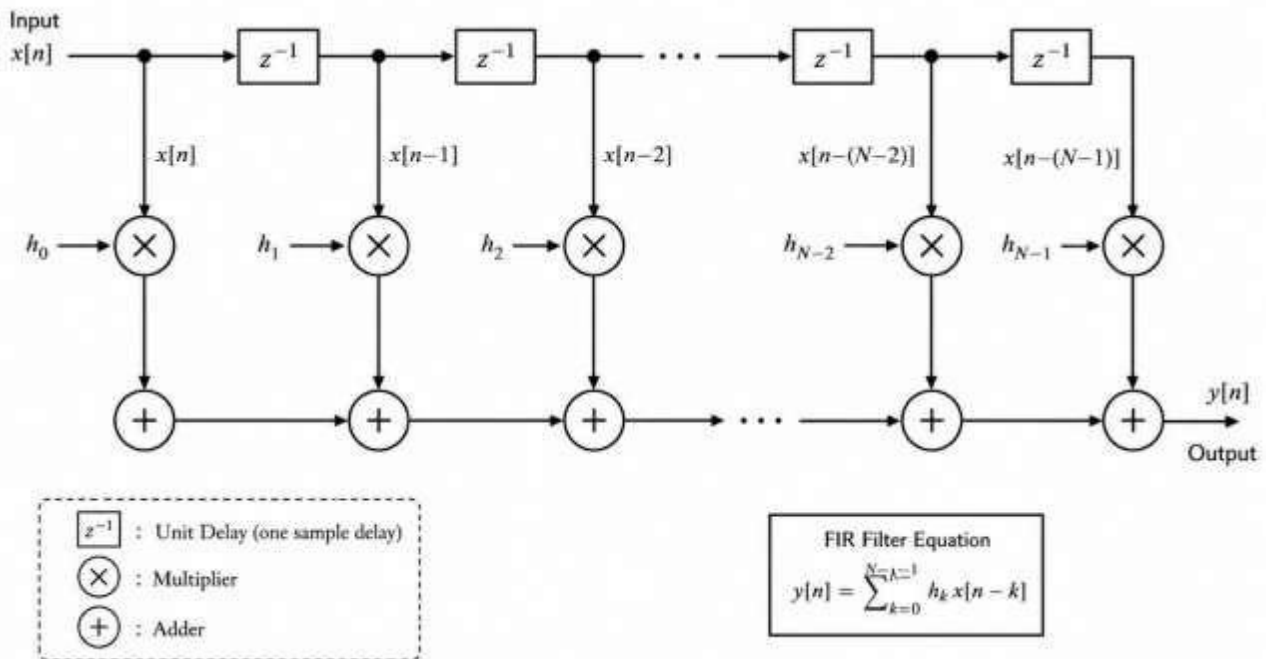
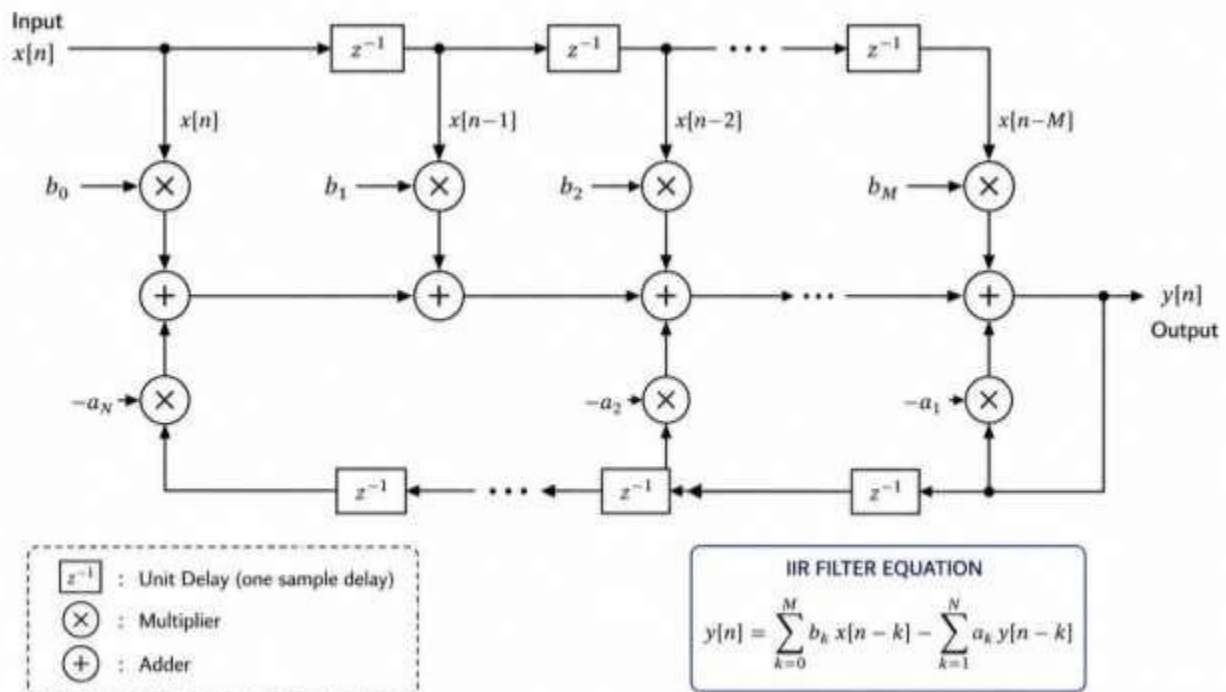
Selected IIR Filter Design

A simple first-order IIR filter can be represented as:

$$y[n] = b_0x[n] + b_1x[n-1] + a_1y[n-1]$$

The IIR filter contains a **feedback path**.

Circuit Diagram:

FIR-Filter Circuit Diagram

IIR-Filter Circuit Diagram


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Verilog Code:

Verilog Code – 4-Tap FIR Filter

```

module fir_filter
( input wire clk,
  input wire reset,
  input wire signed [7:0] x,
  output reg signed [9:0] y
);

  reg signed [7:0] x1;
  reg signed [7:0] x2;
  reg signed [7:0] x3;

  always @(posedge clk) begin
    if (reset) begin
      x1 <= 8'd0;
      x2 <= 8'd0;
      x3 <= 8'd0;
      y <= 10'd0;
    end
    else begin

      // FIR Filter:
      // y[n] = x[n] + x[n-1]
      //       + x[n-2] + x[n-3]

      y <= x + x1 + x2 + x3;

      // Delay line
      x3 <= x2;
      x2 <= x1;
      x1 <= x;

    end
  end
endmodule

```

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Verilog Code – Simple IIR Filter

```

module iir_filter
( input wire clk,
  input wire reset,
  input wire signed [7:0] x,
  output reg signed [9:0] y
);

  reg signed [7:0] x1;
  reg signed [9:0] y1;

  always @(posedge clk) begin
    if (reset) begin
      x1 <= 8'd0;
      y1 <= 10'd0;
      y <= 10'd0;
    end
    else begin

      // Example simplified IIR filter
      y <= x + x1 + (y1 >>> 1);

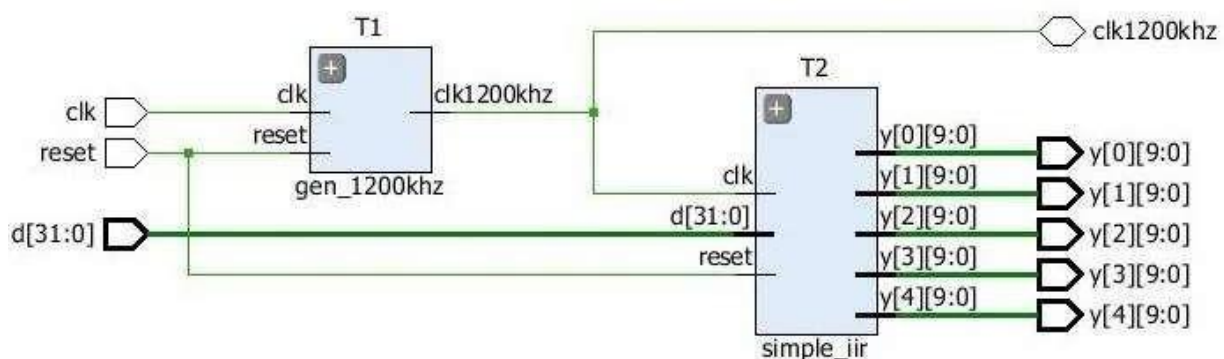
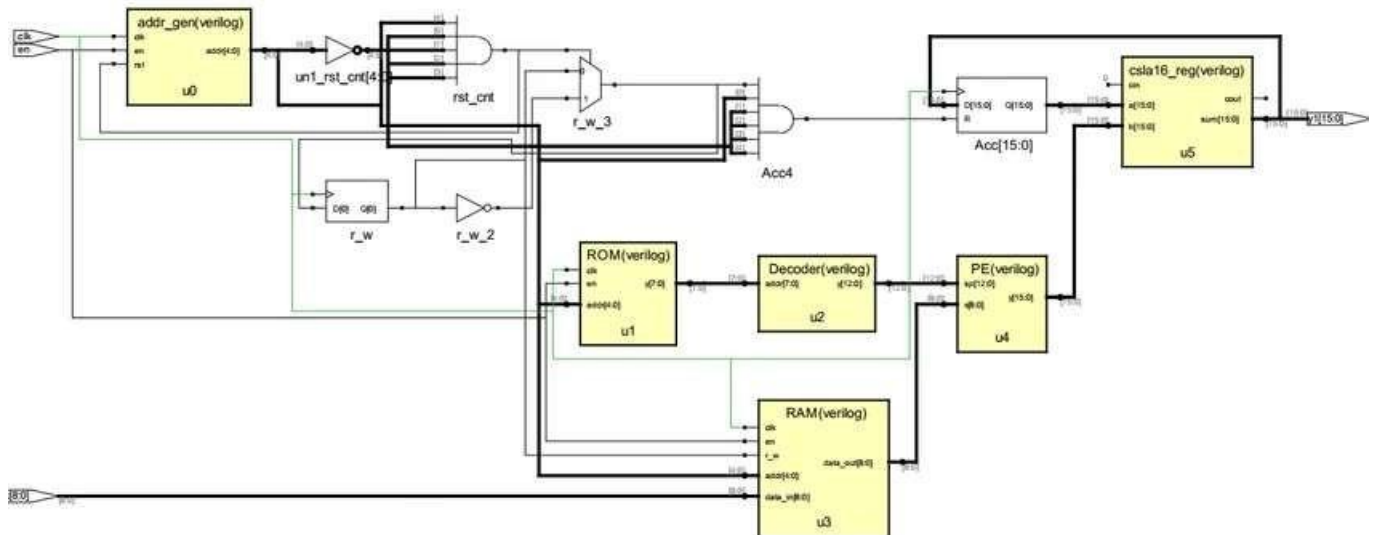
      x1 <= x;
      y1 <= x + x1 + (y1 >>> 1);

    end
  end
endmodule

```

Simulation Results:

Simulation Waveform and RTL Schematic View



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Conclusion:

In this experiment, I studied the design and implementation of FIR and IIR digital filters using Verilog HDL. I understood the difference between FIR filters, which use only feedforward paths, and IIR filters, which use feedback paths. The filter architectures were designed using delay elements, multipliers, and adders. The designs were verified through simulation before hardware implementation. Finally, the filters can be implemented on the Xilinx Artix-7 FPGA (XC7A100T-1CSG324C) hosted on the Digilent Nexys 4 DDR Development Board.

Post Lab Exercise:

1. Why are FIR filters inherently more stable than IIR filters?

FIR filters do not use feedback, so their output does not accumulate indefinitely. Therefore, FIR filters are inherently stable.

2. How does feedback improve the efficiency of an IIR filter?

Feedback allows IIR filters to achieve a desired response using fewer coefficients. This reduces the required hardware resources.

3. Why are FIR filters preferred for applications requiring linear phase response?

FIR filters can easily provide a linear phase response. This prevents phase distortion in the filtered signal.

4. How does increasing the filter order affect frequency selectivity?

Increasing the filter order generally improves frequency selectivity. It provides a sharper transition between passband and stopband.

5. Why is coefficient precision important in digital filter implementation?

Coefficient precision affects the accuracy of the filter response. Low precision can introduce quantization errors.

6. How does FPGA implementation improve digital filter performance?

FPGAs provide parallel processing and high-speed hardware implementation. This improves the throughput of digital filters.

7. Why are multiply-accumulate (MAC) units critical in FIR filter design?

FIR filters require multiplication of samples and coefficients followed by addition. MAC units efficiently perform these operations.

8. How does pipelining enhance filter throughput in programmable hardware?

Pipelining allows multiple filter operations to execute simultaneously in different stages. This increases the processing throughput.

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9. Why is resource sharing used in FPGA-based filter implementation?

Resource sharing reduces the number of multipliers and other hardware components required. This helps save FPGA resources.

10. How does finite word length affect digital filter performance?

Finite word length introduces quantization and rounding errors. This can reduce the accuracy of the filter output.